MD. IMRAN HOSSAIN SHAD

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RESEARCH INTERESTS

- Nanoparticle Synthesis
- Nanocomposites
- Doped Metal-Oxide Semiconductors
- Mechanical Testing
- Materials Characterization
- PFAS Degradation
- Photocatalysis & Environmental Remediation

RESEARCH EXPERIENCE

1. **Graduate Research Assistant** **Jun 2025 – Present**
   Lab: Nano Synthesis Lab
   Department of Nanomaterials and Ceramic Engineering.
   Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh.
   Research Topic: Materials Engineering, Biomaterials, Photocatalysis, PFAS. Currently, I am fabricating metal-free nanocomposites as photocatalysts to enhance the degradation of real-world pollutants like PFAS. This research is funded by the Committee for Advanced Studies and Research (CASR), BUET.

2. **Research Assistant** **Jul 2023 – May 2024**
   Bangladesh Council of Scientific and Industrial Research (BCSIR), Dhaka, Bangladesh.
   - Conducted research and participated in ongoing professional training programs to broaden technical expertise.
   - Used SEM and EDS for surface morphology and composition, XRD for structural analysis, UV–Vis spectroscopy for optical analysis, and thermal analysis.

3. **Undergraduate Research Assistant** **Aug 2022 – Jun 2023**
   Lab: Surface Engineering Lab
   Department of Materials Science & Engineering
   Rajshahi University of Engineering and Technology (RUET), Bangladesh.
   Research Topic: Antibacterial Activities of Zinc Oxide and Tin-Cobalt Co-Doped Zinc Oxide Nanoparticles Synthesized via a Green Route Using *Polyalthia longifolia.*

ACADEMIC CREDENTIALS

1. **M.Sc. in Nanomaterials and Ceramic Engineering** **Apr 2024 – Present**
   Bangladesh University of Engineering & Technology (BUET), Dhaka

   Courses: Advanced Characterization of Glass & Ceramics (A), Ceramic Renewable Energy (A), Bio-Ceramics (A), Phase Equilibria in Glass & Ceramics (B+), Research Methodology & Skills (B+), Structure & Properties of Glass and Ceramics (B+). **(Coursework Completed)**
   Thesis: "Hydrothermal Fabrication of Graphitic Carbon Nitride and Hydroxyapatite Nanocomposite for Enhanced Photocatalytic Degradation of PFAS." **(Ongoing)**

2. **B.Sc. in Materials Science & Engineering** **Jan 2018 – Oct 2023**
   Rajshahi University of Engineering & Technology (RUET), Bangladesh.
   CGPA: 3.10/4.00 (3.45/4.00 in the last 62 credits)

PUBLICATIONS

Journal:

1. **M.I.H. Shad,** M.R. Islam, M.A. Rahaman, A. Molla, H. Rahaman, MS. Islam *"Physicochemical Properties of Cobalt-Tin Co-Doped ZnO Nanoparticles for Efficient Photocatalytic Activity".* (Under Review)
2. A. Molla, **M.I.H. Shad,** S.I. Sohan, S. Ghosh, *"Chitosan-Assisted Synthesis of ZnO and Zn_2TiO_4/ZnO Nanomaterials: Physicochemical properties, Antibacterial Activity and Antibiotic degradation efficacy".* (Under Review)
3. **M.I.H. Shad,** MS. Islam, "Hydrothermal Fabrication of Graphitic Carbon Nitride and Hydroxyapatite Nanocomposite for Enhanced Photocatalytic Degradation of PFAS." (In Preparation)
4. R.Tareq, MMA. Khan, **M.I.H. Shad,** SH. Firoz, *"Strengthening Mechanism in Reversibly pH-responsive Nanocrystalline Cellulose-Reinforced Hydrogel for biomedical application".* (In Preparation)

Conference Papers:

1. **M.I.H. Shad,** M.R. Islam, M.A. Rahaman, H. Rahaman, SMN.Rokon, *"Simultaneous Amelioration in Photocatalytic and Anti-Bacterial Activities of Cobalt & Tin Co-Doped Zinc Oxide Nanoparticles Synthesized Via Green Route Using Polyalthia Longifolia"* Proceedings of the 2nd International Conference on Frontiers in Science (ICFS), Dhaka, Bangladesh, Jan 15-16, 2026. (Awarded the best poster presentation)
2. R.Tareq, MMA. Khan, **M.I.H. Shad,** SH. Firoz *"Functionalized Nanocrystalline Cellulose in improving Tensile Strength of Hyperelastic Poly (Acrylic acid-co-Acrylamide) Nanocomposite Hydrogel"*, Proceedings of the 2nd International Conference on Frontiers in Science (ICFS), Dhaka, Bangladesh, Jan 15-16, 2026.

TECHNICAL PRESENTATIONS

1. **Poster Presentation, Materials Symposium 2025** **Feb 2025**
   Materials Research Centre, BUET, Dhaka, Bangladesh
   Topic: "Antibacterial Activities of Pure Zinc Oxide and Tin–Cobalt Co-doped Zinc Oxide Nanoparticles Synthesized via a Green Route Using *Polyalthia longifolia*." (Awarded 5th Place)

TECHNICAL AND INSTRUMENTATION SKILLS

Nano-synthesis, Photocatalytic Activity Testing (UV–Vis Photocatalysis System), X-ray Diffraction (XRD), Scanning Electron Microscopy (SEM), Energy Dispersive X-ray Spectroscopy (EDS), UV–Visible Spectroscopy (UV–Vis), Optical Emission Spectroscopy (OES), Fourier Transform Infrared Spectroscopy (FTIR), Differential Scanning Calorimetry (DSC), Thermogravimetric Analysis (TGA), Liquid Chromatography–Mass Spectrometry (LC-MS), Metallography, Failure Analysis.

SOFTWARE AND PROGRAMMING

Characterization:	OriginPro, X'Pert HighScore Plus, ImageJ
Programming:	C, MATLAB, LaTeX, HTML, MySQL
Engineering:	AutoCAD
Graphics:	Adobe Illustrator, Adobe Photoshop
Office:	MS Word, MS Excel, MS PowerPoint, Google Sheets

PROFESSIONAL TRAINING

1. Position: Intern (industrial attachment) **Apr 2022**
Bangladesh Industrial Technical Assistance Centre (BITAC)
Tejgaon, Dhaka, Bangladesh

- Gained knowledge of quality control and management through OES and metallographic testing.
- Gained exposure to foundry, pattern shop, and furnace operations.
- Observed metal fabrication processes using heavy machinery.

AWARDS/ PROJECTS/ SCHOLARSHIPS

- Best poster presenter, 2nd ICFS: ITGI 2026, organized by Faculty of Science, BUET **Jan 2026**
- 5th Place, Materials Symposium 2025, organized by the Materials Research Centre, BUET **Feb 2025**
- Design of a Ball Mill Grinder (Lab Project) **Jul 2023**
- Failure and Artefact Analysis of a Diesel Engine Crankshaft **Jul 2023**
- Failure Analysis of a Hydraulic Press Trunk Piston (Surface Engineering Lab, RUET) **Jun 2023**
- Technical Scholarship, Rajshahi University of Engineering & Technology (RUET)
- Board Scholarships, Government of Bangladesh

REFERENCES

Dr. Md. Saiful Islam
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Bangladesh University of Engineering
and Technology (BUET), Dhaka
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S. M. Nasim Rokon
Assistant Professor
Materials Science & Engineering
Rajshahi University of Engineering
and Technology (RUET), Rajshahi
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